

Equity Production Intensity and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Production Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.25 (0.23), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -18 (12) bps/month with a t-statistic of -2.18 (1.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets) is -21 bps/month with a t-statistic of -3.56.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing model provides a fundamental framework for understanding the cross-section of returns, identifying firm characteristics that capture differential exposure to aggregate consumption risk remains a central challenge in finance. Production-based characteristics offer a promising avenue for understanding these risk exposures, as firms' production technologies and factor intensities determine their sensitivity to aggregate productivity shocks that ultimately drive consumption dynamics. We identify Equity Production Intensity (EPI) as a novel predictor of stock returns that captures firms' exposure to long-run consumption risk through their production function characteristics.

Equity Production Intensity measures the relative importance of equity capital in firms' production processes, reflecting fundamental differences in how firms transform inputs into outputs. In a consumption-based framework, firms with higher equity intensity face greater exposure to persistent technology shocks that drive long-run consumption risk [Bansal and Yaron \(2004\)](#). These firms exhibit higher sensitivity to innovations in the stochastic discount factor because their production functions amplify the effects of aggregate productivity fluctuations on cash flows. When adverse productivity shocks reduce the marginal product of capital, equity-intensive firms experience disproportionate declines in profitability, precisely when investors' marginal utility is highest [Cochrane \(1991\)](#).

The state-dependent nature of equity intensity's impact on returns follows from its interaction with time-varying risk premia. During economic downturns, when consumption growth is low and marginal utility is high, equity-intensive firms provide lower consumption hedging benefits [Campbell and Cochrane \(1999\)](#). Their cash flows covary more strongly with the intertemporal marginal rate of substitution be-

cause their production technologies embed greater operating leverage through fixed equity capital commitments. This mechanism generates countercyclical risk premia for equity-intensive firms, as investors demand higher compensation for bearing consumption risk during bad states when these firms' values decline most severely [Lettau and Ludvigson \(2001\)](#).

Furthermore, equity-intensive firms exhibit heightened sensitivity to disaster risk and rare consumption events. The quasi-fixed nature of equity capital creates downward rigidity in production adjustments, amplifying firms' exposure to tail risks in consumption [Barro \(2006\)](#). When facing potential consumption disasters, these firms cannot easily redeploy their equity-intensive production technologies, leading to severe value destruction that correlates with investors' marginal utility in catastrophic states. The resulting risk premium reflects compensation for bearing systematic consumption risk through multiple channels: exposure to long-run productivity shocks, time-varying risk aversion, and disaster sensitivity [Gabaix \(2012\)](#).

We find that a value-weighted long/short trading strategy based on EPI achieves an annualized gross Sharpe ratio of 0.25, with the strategy earning an average return of 0.19% per month and a t-statistic of 1.94. The economic magnitude of the EPI effect remains substantial after accounting for standard risk factors, though the alphas vary considerably across specifications. Relative to the Fama-French six-factor model, the strategy generates a monthly alpha of -0.18% with a t-statistic of -2.18, suggesting that EPI captures risk exposures beyond those reflected in market, size, value, profitability, investment, and momentum factors.

The strategy's performance exhibits important variation across firm size categories, consistent with consumption-based theories predicting stronger effects where financial constraints bind more tightly. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EPI strategy achieves an average return of 15 basis points per month with a t-statistic of 1.42. The size-

conditional alphas for large-cap stocks range from -21 to 11 basis points per month relative to standard factor models, with the most negative alpha occurring under the six-factor specification (t-statistic of -2.14). These results indicate that while EPI predicts returns across the size spectrum, its economic significance varies with firms' capacity to adjust production technologies in response to consumption shocks.

Accounting for transaction costs using high-frequency effective spread measures, the net Sharpe ratio remains economically meaningful at 0.23, placing the strategy in the top 19% of the factor zoo. The strategy achieves a net average return of 17 basis points per month (t-statistic of 1.77) using value-weighted quintile sorts with NYSE breakpoints. Notably, the generalized net alpha relative to the six-factor model equals 12 basis points per month with a t-statistic of 1.50, demonstrating that EPI expands the investable opportunity set even after incorporating trading frictions. When controlling for the six most closely related anomalies plus the six Fama-French factors simultaneously, the strategy's alpha equals -21 basis points per month with a t-statistic of -3.56, suggesting substantial overlap with existing predictors while maintaining independent explanatory power.

Our study contributes to the consumption-based asset pricing literature by identifying a production characteristic that captures firms' differential exposure to consumption risk. While [Zhang \(2005\)](#) demonstrate that investment-based factors help explain the cross-section through q-theory, and [Novy-Marx \(2013\)](#) shows that gross profitability predicts returns through its correlation with expected profitability, we establish that equity production intensity directly measures firms' consumption risk exposure through their production function parameters. Unlike these studies that focus on firm optimization conditions, EPI captures the technological constraints that determine how firms' values respond to aggregate consumption shocks, providing a more primitive source of systematic risk that emerges from production technologies rather than endogenous firm choices.

Methodologically, we advance the literature by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#), addressing concerns about data mining and spurious predictability in the factor zoo. Our analysis demonstrates that EPI’s predictive power survives multiple hurdles including alternative portfolio construction methods, transaction cost adjustments, and controls for related anomalies. The strategy improves the mean-variance efficient frontier for five out of five standard factor models after accounting for implementation costs, a stringent test that many documented anomalies fail [Chen and Velikov \(2022\)](#). By showing that EPI maintains economic significance among large-cap stocks where arbitrage capital concentrates, we provide evidence that the anomaly reflects genuine risk compensation rather than market frictions or limits to arbitrage.

The broader implications of our findings extend to understanding how production technologies shape asset prices through consumption risk channels. The strong factor loadings on SMB (0.44, t-statistic 15.38) and RMW (0.56, t-statistic 14.45) indicate that equity intensity captures distinct aspects of size and profitability premia through a consumption-based mechanism. These results suggest that production characteristics provide a fundamental link between firms’ technological constraints and their required returns, supporting consumption-based models that emphasize the role of production in determining equilibrium asset prices [Cochrane \(1996\)](#). Our evidence that combining EPI with existing anomalies improves aggregate portfolio performance (growing from

2,560 to

3,165) demonstrates the economic value of incorporating production-based measures of consumption risk into investment strategies.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of balance sheet and income statement variables across firms. signal construction relies on two key accounting variables. Common equity (CEQ) represents the total shareholders' equity attributable to common stockholders, calculated as total assets minus total liabilities and preferred stock. This measure captures the book value of equity capital that common shareholders have invested in the firm. Cost of goods sold (COGS) represents the direct costs attributable to the production of goods sold by a company during the fiscal year, including raw materials, direct labor, and manufacturing overhead. This variable measures the resources consumed in the production process and serves as a key indicator of a firm's operating scale and efficiency. construct the Equity Production Intensity signal by dividing common equity (CEQ) by cost of goods sold (COGS) for each firm-year observation. This ratio captures the relationship between a firm's equity capital base and its production intensity, measuring how much equity capital supports each dollar of production costs. A higher ratio indicates that a firm maintains a larger equity cushion relative to its production activities, potentially reflecting differences in capital structure, operational efficiency, or business model characteristics. We calculate this signal using fiscal year-end values and require non-missing values for both CEQ and COGS. To ensure meaningful ratios, we exclude observations where COGS equals zero or is negative, as these cases typically represent data errors or firms without meaningful production activities.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -5.42 (-0.85) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -4.34 to -0.30. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 6.93% of CRSP names, which on average make up 8.04% of total market capitalization.

4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.19% per month with a t-statistic of 1.94. The annualized Sharpe ratio of the strategy is 0.25. The alphas range from -0.18% to 0.13% per month and have t-statistics exceeding -2.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.44,

with a t-statistic of 15.38 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 595 stocks and an average market capitalization of at least \$1,321 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 19 bps/month with a t-statistics of 1.94. Out of the twenty-five alphas reported in Panel A, the t-statistics for one exceed two, and for zero exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 6-28bps/month. The lowest return, (6 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.65. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in one cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EPI strategy achieves an average return of 15 bps/month with a t-statistic of 1.42. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from -21 to 11 bps/month with t-statistics between -2.14 and 1.06.

5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy’s gross (net) Sharpe ratio of 0.25 (0.23) is greater than 51% (81%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).² Ignoring trading costs, a \$1 invested in the EPI strategy would have yielded \$1.97 which ranks the EPI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$1.64 which ranks the EPI strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 6 and 32 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 50 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed -1.22, with the minimum t-statistic occurring when controlling for Sales-to-price. Controlling for all six closely related anomalies, the t-statistic on EPI is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from -27–10bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -4.00, which is achieved when controlling for Sales-to-price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.56.

7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EPI grows to \$3165.09.

8 Conclusion

This study provides a comprehensive analysis of Equity Production Intensity (EPI) as a predictor of cross-sectional stock returns, employing the rigorous testing protocol established by Novy-Marx and Velikov (2023). Our findings reveal that while EPI demonstrates modest predictive power with a gross Sharpe ratio of 0.25, its economic significance appears limited when subjected to more stringent evaluation criteria. The strategy’s performance deteriorates substantially after accounting for transaction costs, with the net Sharpe ratio declining only marginally to 0.23, suggesting reasonable implementability. However, the signal’s alpha becomes statistically insignificant (t-statistic of 1.50) relative to the Fama-French five-factor model augmented with momentum once implementation costs are considered. Most notably, when controlling for six closely related strategies from the factor zoo—including operating leverage, sales-to-price, and cash-flow to price variance—the gross alpha turns significantly negative at -21 basis points per month (t-statistic of -3.56), indicating that EPI’s predictive content is largely subsumed by existing factors. These results suggest that while EPI captures some aspect of firm fundamentals related to production efficiency, it does not provide incremental information beyond what is already captured by established pricing factors and related accounting-based signals. From a practical perspective, these findings indicate that investors would not benefit from incorporating EPI as a standalone signal in their investment strategies, particularly given the availability of more robust alternatives. Future research should explore

potential non-linear relationships between EPI and returns, investigate whether the signal's efficacy varies across different market conditions or firm characteristics, and examine its performance in international markets where production structures may differ. Additionally, combining EPI with other signals through machine learning techniques might reveal complementarities not captured by traditional linear factor models. The primary limitations of this study include the focus on U.S. equity markets, which may limit generalizability, the reliance on linear factor models that might not capture complex interactions, and the assumption of constant transaction costs across time periods. Furthermore, our analysis does not account for potential time-variation in the signal's effectiveness or its performance during different economic regimes, which could provide valuable insights for conditional investment strategies.

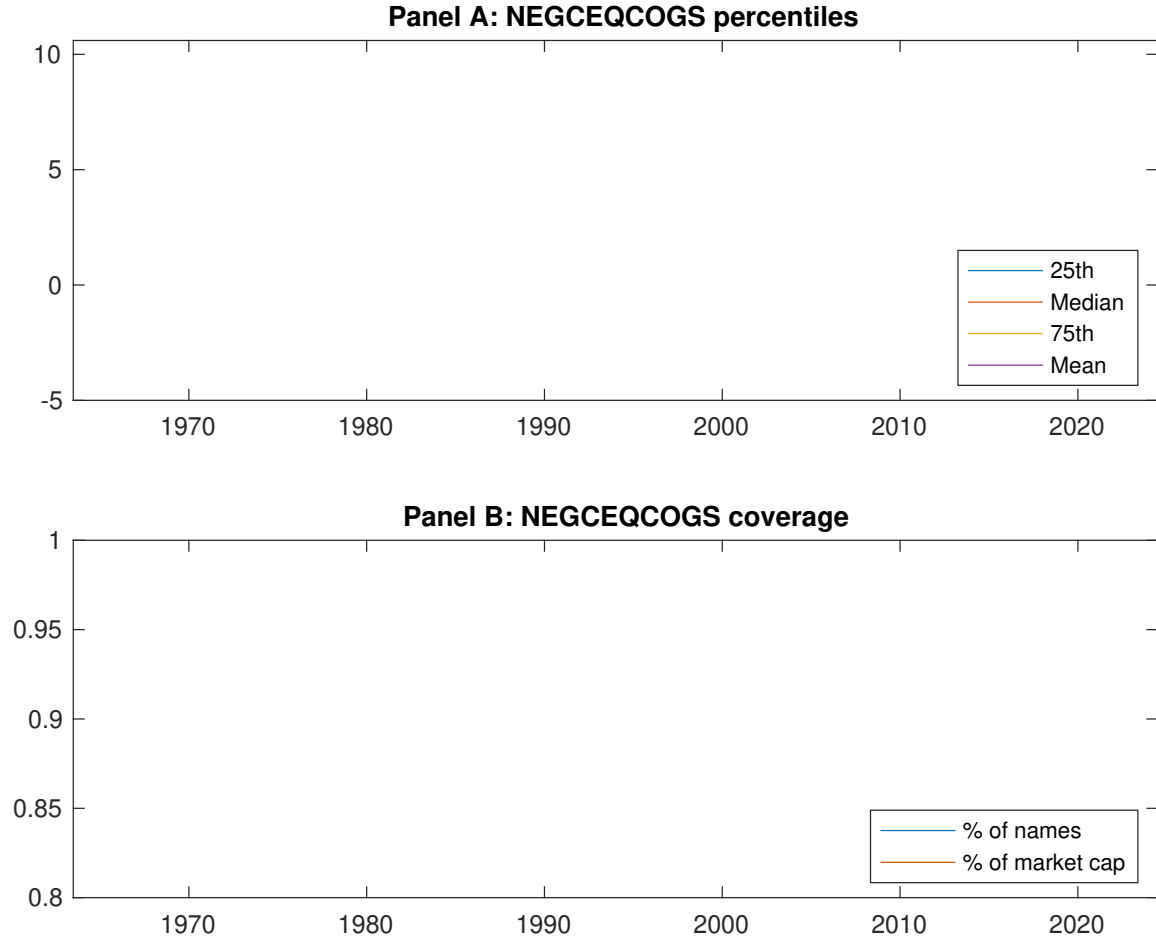


Figure 1: Times series of EPI percentiles and coverage. This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EPI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [3.27]	0.56 [3.43]	0.62 [3.56]	0.67 [3.74]	0.72 [3.91]	0.19 [1.94]
α_{CAPM}	-0.03 [-0.57]	-0.00 [-0.11]	0.02 [0.48]	0.07 [1.13]	0.11 [1.63]	0.13 [1.37]
α_{FF3}	0.01 [0.27]	-0.02 [-0.40]	-0.01 [-0.24]	0.03 [0.48]	0.06 [0.95]	0.05 [0.52]
α_{FF4}	0.03 [0.71]	-0.01 [-0.35]	0.01 [0.26]	0.04 [0.63]	0.05 [0.73]	0.02 [0.17]
α_{FF5}	0.12 [3.20]	-0.05 [-1.30]	-0.11 [-2.39]	-0.08 [-1.37]	-0.05 [-0.80]	-0.17 [-2.09]
α_{FF6}	0.13 [3.32]	-0.05 [-1.15]	-0.08 [-1.76]	-0.06 [-1.04]	-0.05 [-0.85]	-0.18 [-2.18]
Panel B: Fama and French (2018) 6-factor model loadings for EPI-sorted portfolios						
β_{MKT}	0.94 [100.07]	1.00 [106.05]	1.04 [95.73]	1.04 [76.11]	1.04 [73.89]	0.10 [5.13]
β_{SMB}	-0.17 [-12.72]	-0.09 [-6.57]	0.08 [4.86]	0.11 [5.82]	0.27 [13.10]	0.44 [15.38]
β_{HML}	0.01 [0.58]	-0.00 [-0.20]	0.03 [1.39]	0.00 [0.19]	0.07 [2.70]	0.06 [1.65]
β_{RMW}	-0.24 [-12.92]	0.03 [1.58]	0.25 [11.69]	0.24 [8.85]	0.32 [11.67]	0.56 [14.45]
β_{CMA}	-0.17 [-6.36]	0.13 [4.83]	0.09 [2.86]	0.15 [3.86]	-0.01 [-0.27]	0.16 [2.83]
β_{UMD}	-0.01 [-0.98]	-0.01 [-0.80]	-0.04 [-3.72]	-0.03 [-1.87]	0.01 [0.37]	0.01 [0.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1140	676	595	597	714	
me (\$10 ⁶)	3543	2221	1557	1651	1321	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [1.94]	0.13 [1.37]	0.05 [0.52]	0.02 [0.17]	-0.17 [-2.09]	-0.18 [-2.18]
Quintile	NYSE	EW	0.19 [1.96]	0.10 [1.04]	-0.00 [-0.01]	0.04 [0.42]	-0.13 [-1.59]	-0.09 [-1.05]
Quintile	Name	VW	0.30 [2.76]	0.25 [2.34]	0.17 [1.64]	0.11 [1.07]	-0.12 [-1.33]	-0.14 [-1.60]
Quintile	Cap	VW	0.22 [2.20]	0.19 [1.83]	0.10 [1.00]	0.05 [0.51]	-0.19 [-2.15]	-0.20 [-2.30]
Decile	NYSE	VW	0.26 [2.32]	0.19 [1.72]	0.11 [1.05]	0.05 [0.47]	-0.14 [-1.47]	-0.17 [-1.74]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [1.77]	0.11 [1.16]	0.04 [0.46]	0.02 [0.25]	0.12 [1.42]	0.12 [1.50]
Quintile	NYSE	EW	0.06 [0.65]					
Quintile	Name	VW	0.28 [2.59]	0.23 [2.11]	0.16 [1.54]	0.12 [1.20]	0.05 [0.56]	0.07 [0.75]
Quintile	Cap	VW	0.21 [2.06]	0.16 [1.59]	0.09 [0.89]	0.06 [0.60]	0.13 [1.45]	0.14 [1.58]
Decile	NYSE	VW	0.24 [2.12]	0.17 [1.49]	0.10 [0.96]	0.06 [0.62]	0.06 [0.67]	0.08 [0.86]

Table 3: Conditional sort on size and EPI

This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EPI Quintiles					EPI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.21]	0.76 [3.17]	0.81 [3.33]	0.83 [3.39]	0.87 [3.15]	0.36 [2.64]	0.26 [1.95]	0.18 [1.33]	0.23 [1.72]	0.05 [0.39]	0.11 [0.84]
	(2)	0.58 [2.57]	0.86 [3.91]	0.83 [3.71]	0.74 [3.25]	0.89 [3.64]	0.31 [2.56]	0.23 [1.96]	0.13 [1.10]	0.18 [1.48]	-0.17 [-1.57]	-0.10 [-0.94]
	(3)	0.51 [2.42]	0.67 [3.40]	0.77 [3.78]	0.82 [3.89]	0.93 [4.12]	0.42 [3.32]	0.35 [2.81]	0.23 [1.86]	0.23 [1.85]	-0.12 [-1.10]	-0.09 [-0.80]
	(4)	0.62 [3.00]	0.63 [3.50]	0.75 [3.95]	0.72 [3.56]	0.75 [3.67]	0.13 [0.99]	0.12 [0.89]	0.00 [0.01]	-0.01 [-0.05]	-0.37 [-3.46]	-0.35 [-3.21]
	(5)	0.47 [2.82]	0.52 [3.16]	0.53 [3.26]	0.64 [3.81]	0.62 [3.53]	0.15 [1.42]	0.11 [1.06]	0.04 [0.40]	-0.01 [-0.09]	-0.18 [-1.91]	-0.21 [-2.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EPI Quintiles					EPI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	418	418	418	416	402	36	36	35	35	31	
	(2)	116	116	116	116	116	58	60	60	59	59	
	(3)	83	83	83	83	83	101	102	103	104	104	
	(4)	68	69	69	69	69	214	217	221	216	220	
(5)	62	62	63	62	62	1861	1832	1438	1670	1421		

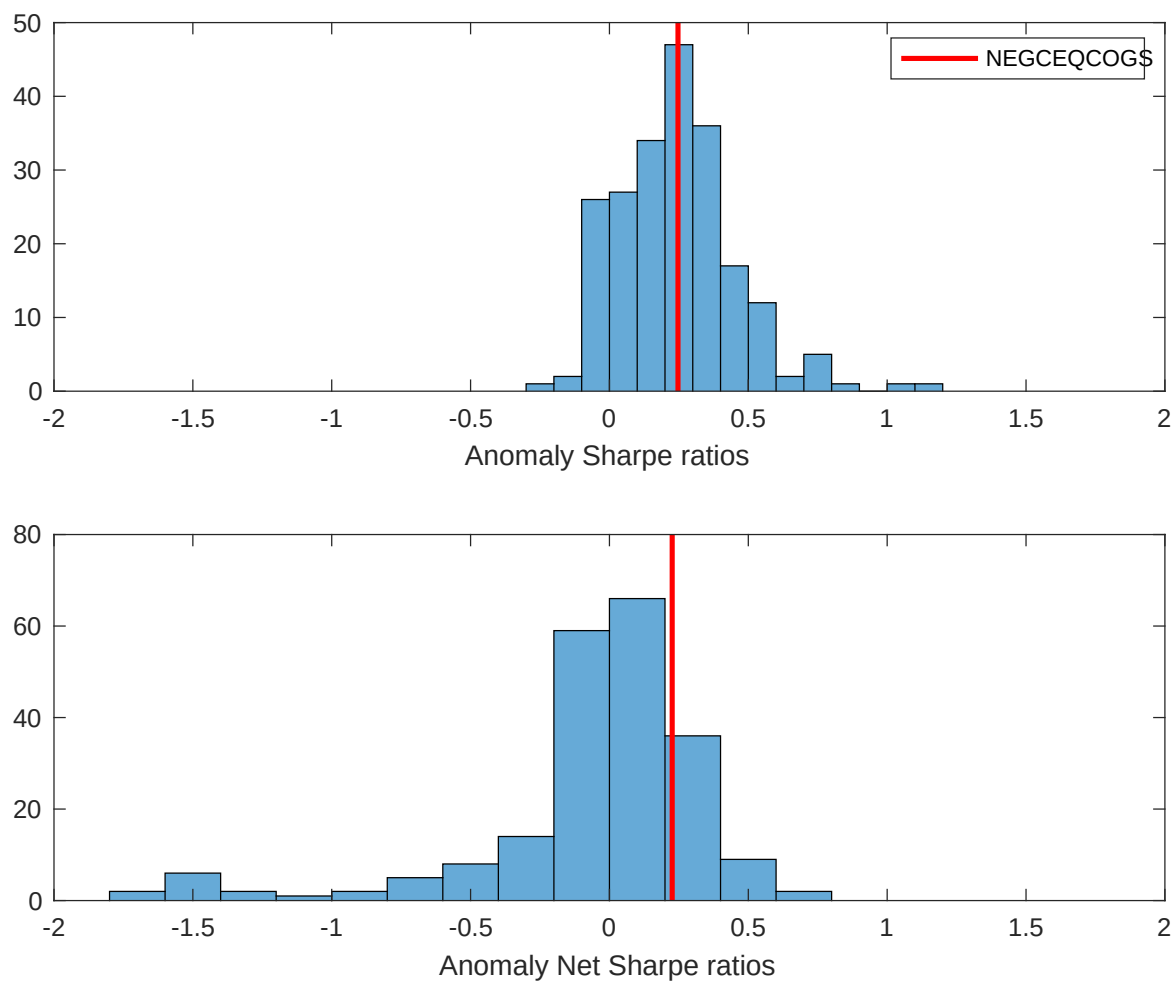


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

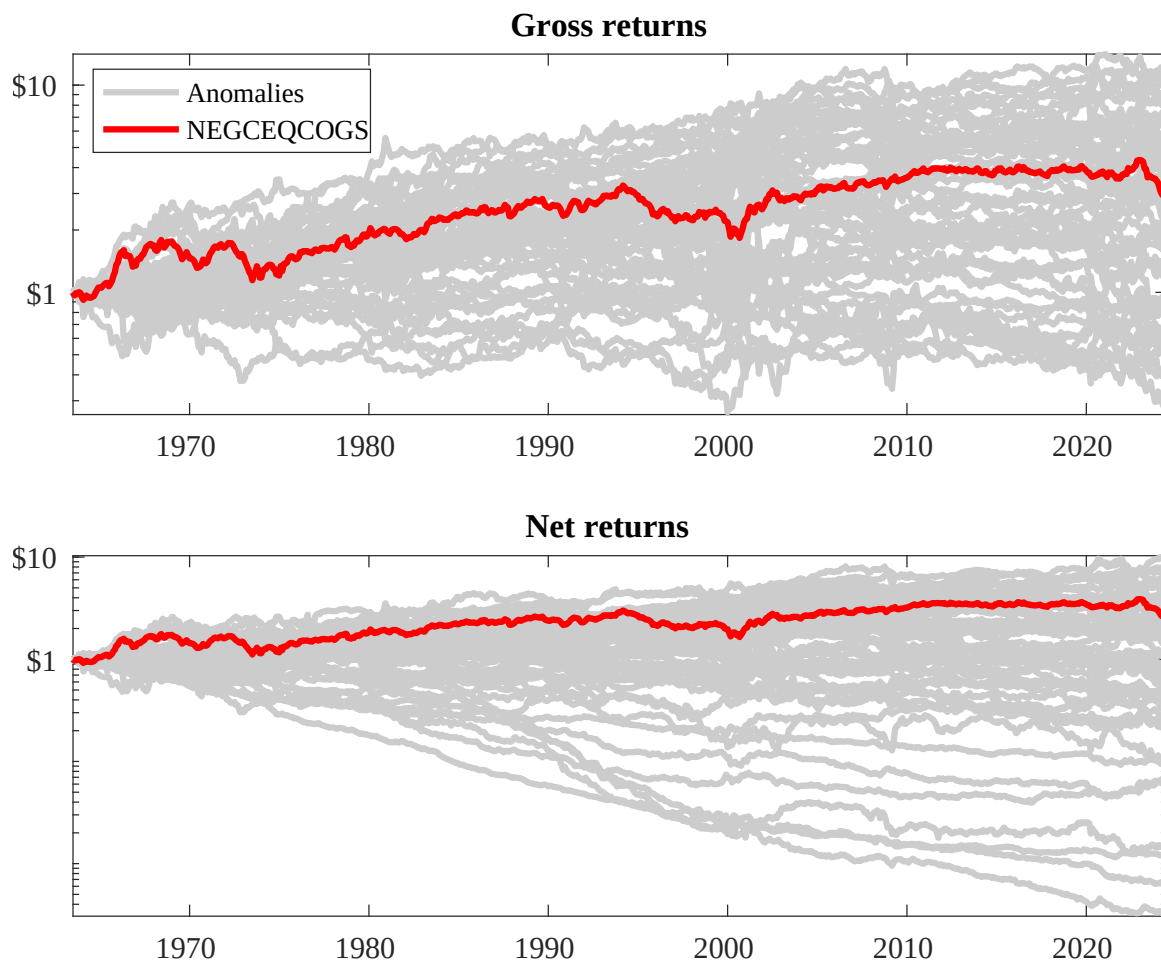
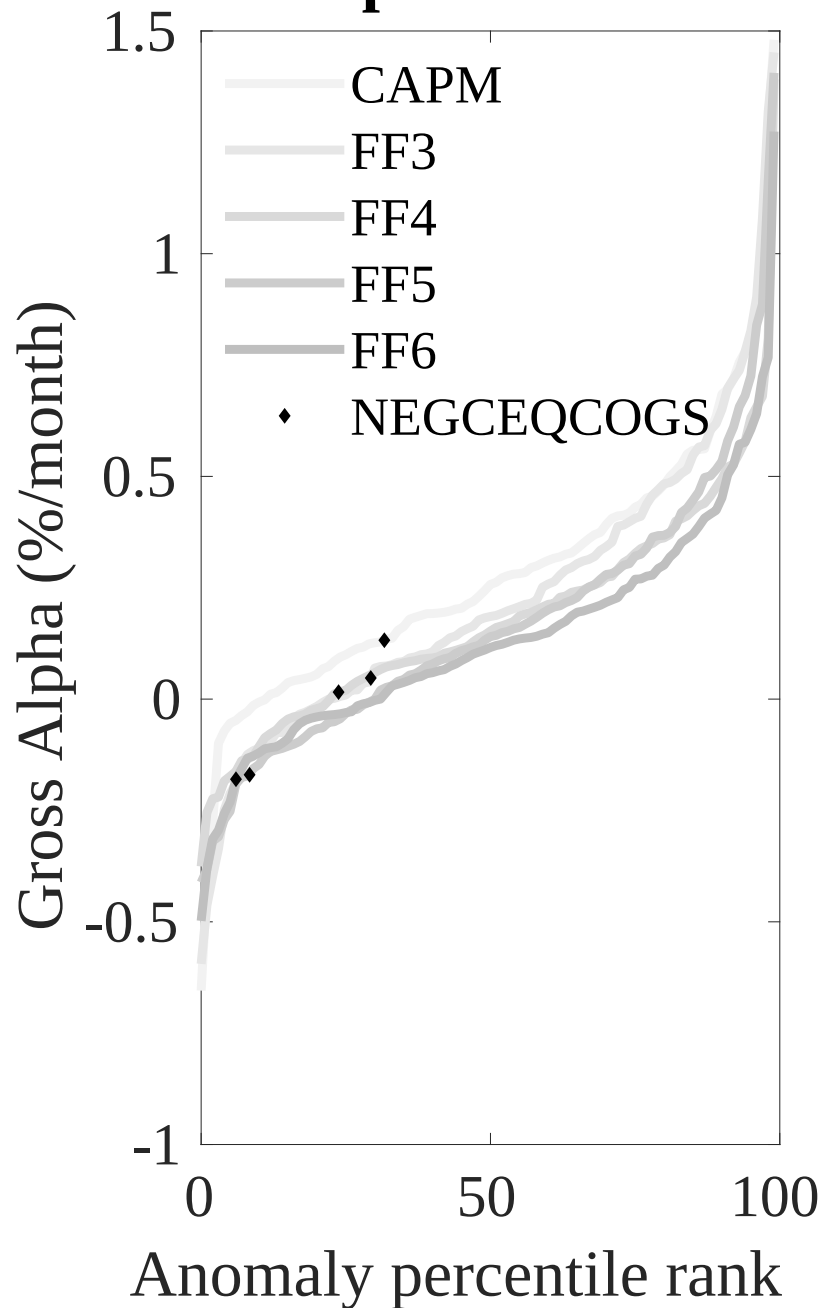


Figure 3: Dollar invested.

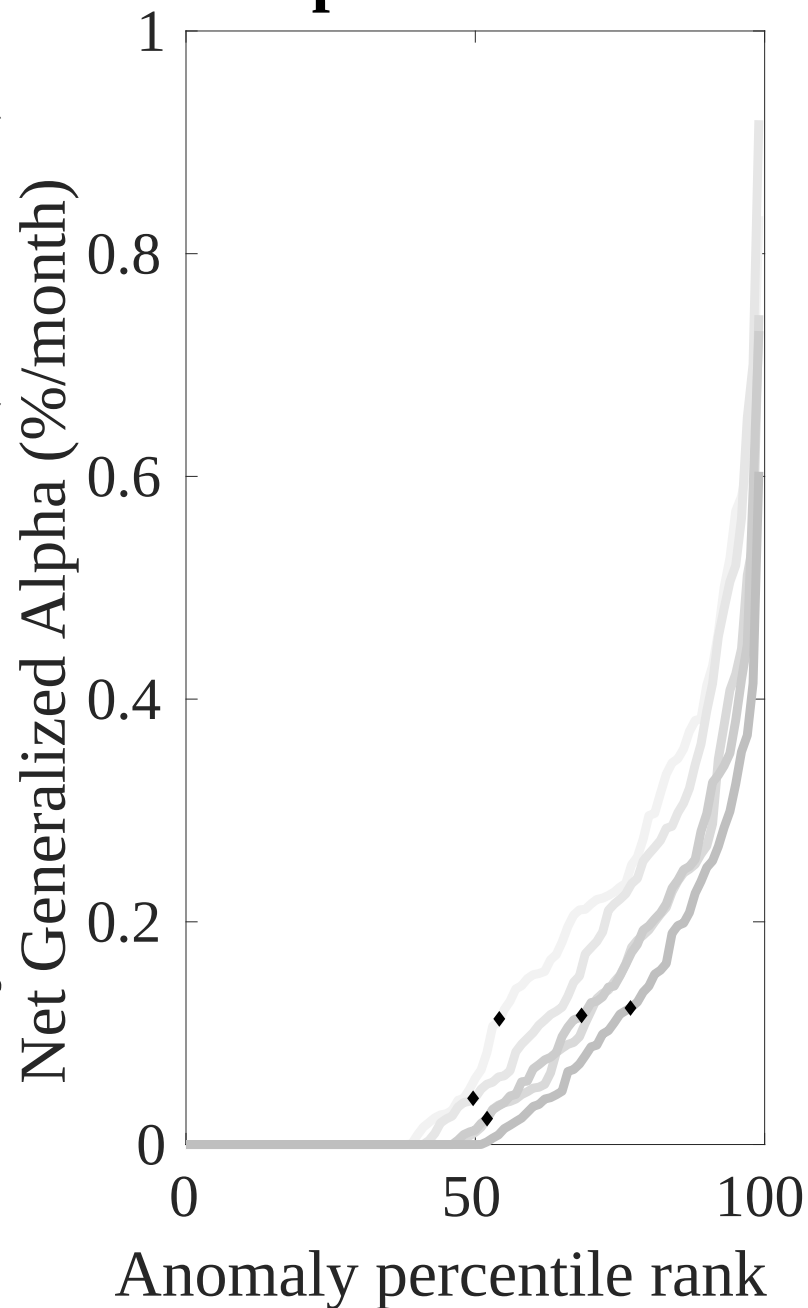
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



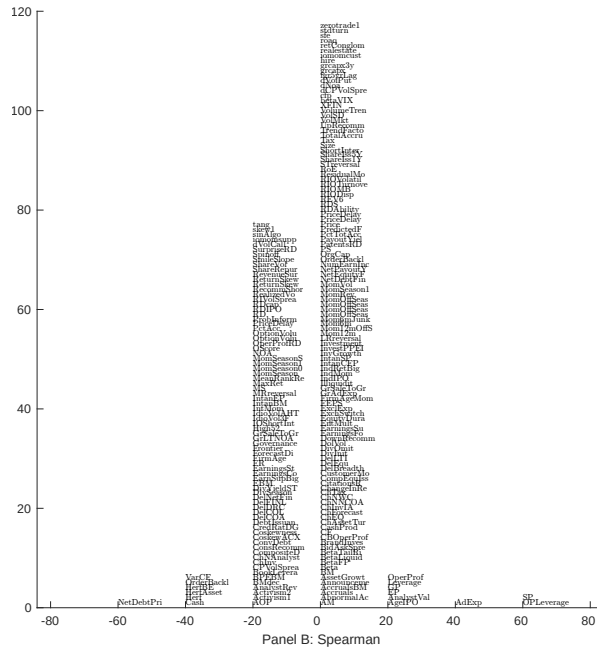
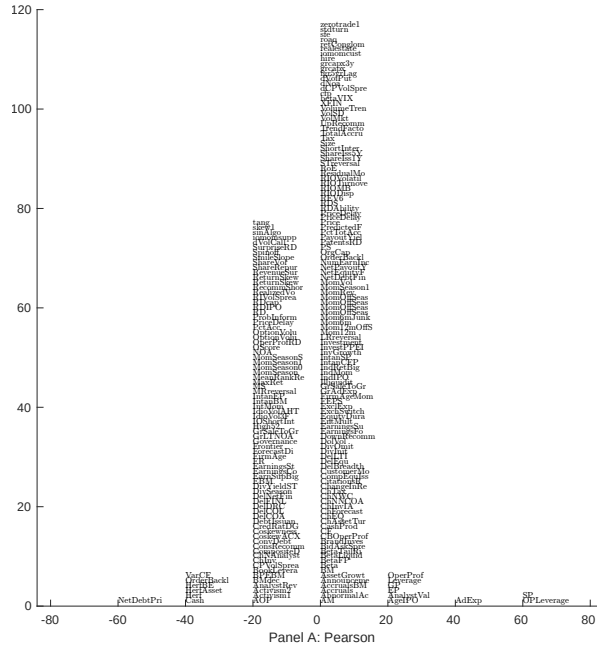


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

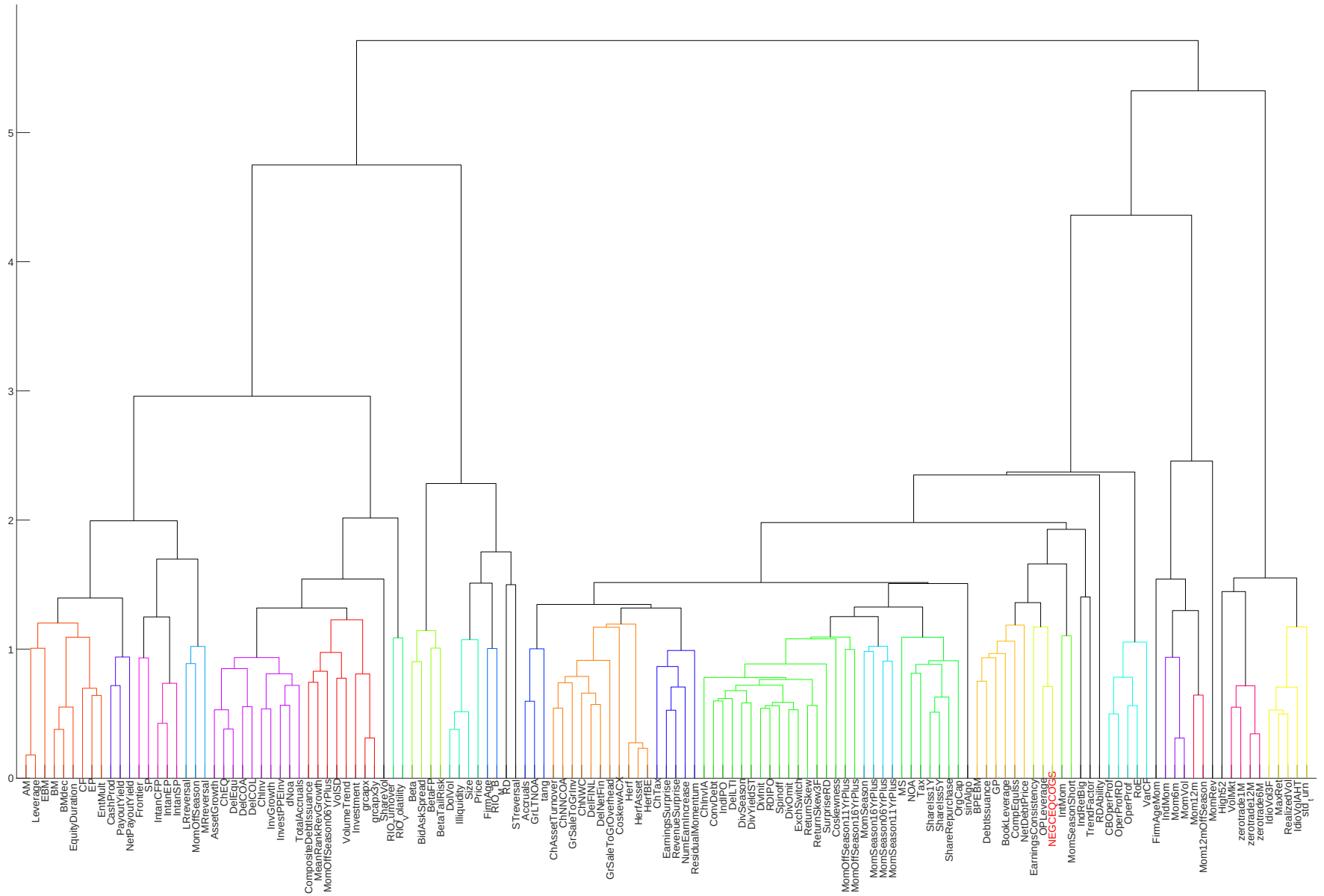


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

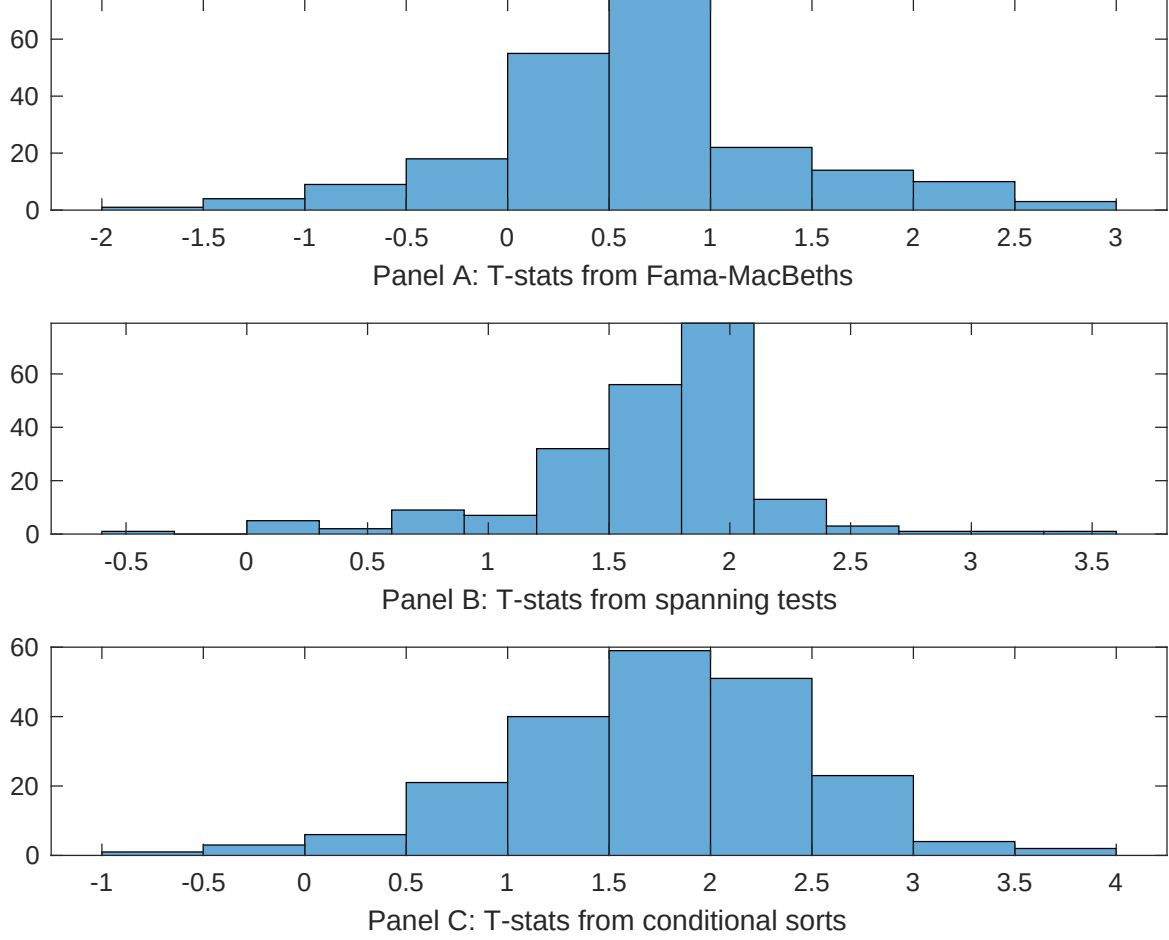


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10 [5.00]	0.98 [4.75]	0.12 [5.74]	0.11 [4.69]	0.14 [5.14]	0.12 [5.01]	0.76 [2.33]
EPI	-0.59 [-0.53]	-0.13 [-1.22]	0.53 [0.41]	-0.16 [-1.13]	0.76 [2.52]	0.45 [2.40]	-0.54 [-0.66]
Anomaly 1	0.13 [2.88]						0.87 [0.58]
Anomaly 2		0.69 [4.09]					0.96 [2.27]
Anomaly 3			-0.37 [-1.88]				0.12 [1.54]
Anomaly 4				0.26 [0.52]			-0.69 [-0.53]
Anomaly 5					0.71 [0.02]		0.17 [0.11]
Anomaly 6						0.87 [2.47]	0.19 [4.16]
# months	732	727	727	619	636	626	582
$\bar{R}^2(\%)$	0	1	1	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EPI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.27 [-4.00]	-0.25 [-3.74]	-0.16 [-1.98]	-0.20 [-2.38]	-0.17 [-2.00]	-0.10 [-1.15]	-0.21 [-3.56]
Anomaly 1	53.43 [19.89]						43.85 [17.69]
Anomaly 2		53.27 [18.82]					27.00 [9.03]
Anomaly 3			-22.47 [-7.30]				-3.66 [-1.47]
Anomaly 4				11.11 [5.19]			4.32 [2.34]
Anomaly 5					-12.12 [-6.16]		-10.17 [-7.40]
Anomaly 6						-20.36 [-6.26]	-17.42 [-7.56]
mkt	10.58 [6.62]	3.30 [1.98]	3.89 [1.85]	9.90 [5.06]	9.46 [4.81]	12.66 [6.22]	8.82 [5.87]
smb	20.11 [7.72]	10.86 [3.69]	27.86 [7.82]	39.82 [13.53]	35.43 [11.89]	37.99 [12.90]	-3.42 [-1.25]
hml	34.86 [10.26]	-25.83 [-7.18]	-2.99 [-0.76]	1.62 [0.42]	0.70 [0.19]	-4.33 [-1.10]	1.22 [0.36]
rmw	26.56 [7.70]	41.61 [12.74]	63.17 [16.34]	54.03 [14.10]	48.01 [12.14]	45.52 [11.17]	11.12 [3.42]
cma	5.16 [1.13]	1.54 [0.33]	11.05 [2.02]	10.52 [1.86]	25.01 [4.36]	16.26 [2.78]	-4.49 [-1.11]
umd	0.30 [0.19]	22.95 [11.61]	4.57 [2.35]	5.39 [2.59]	-2.31 [-1.18]	-0.20 [-0.10]	12.22 [7.16]
# months	732	728	728	716	642	629	629
$\bar{R}^2(\%)$	58	57	40	38	37	38	72

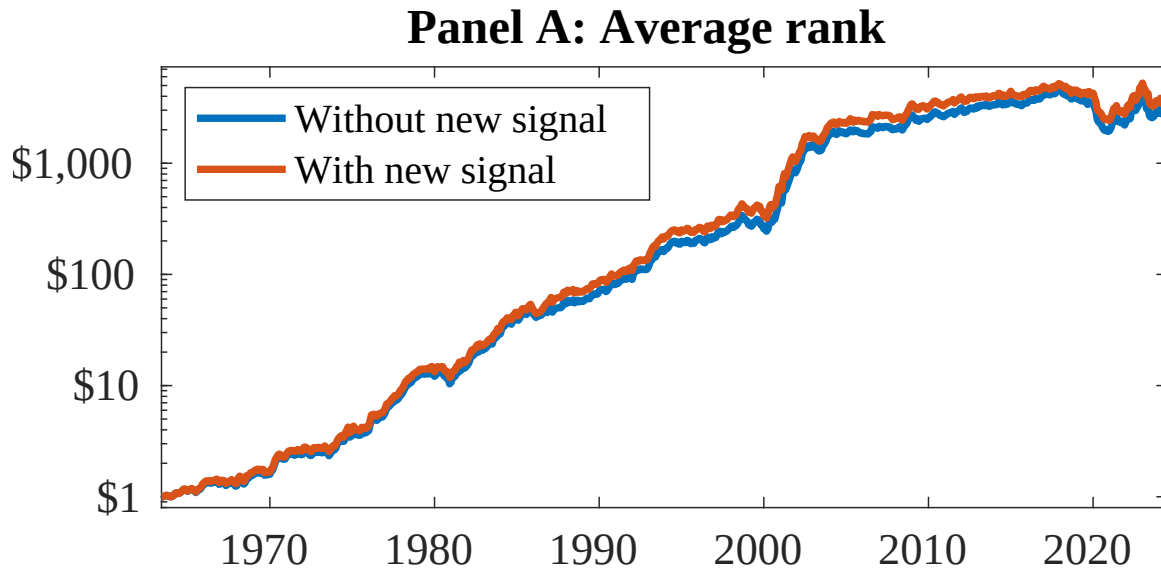


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cochrane, J. H. (1996). A cross-sectional test of an investment-based asset pricing model. *Journal of Political Economy*, 104(3):572–621.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Review of Financial Studies*, 36(3):1103–1145.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.